

RESEARCH INTERESTS

- Formal language theory: Parikh matrices and language hierarchies
- Neuro-symbolic methods for NLP: grammar-based data augmentation, few-shot learning, and explainable AI
- Computational linguistics and code intelligence: complexity analysis and code summarization

RESEARCH STATEMENT

- **Integrating Formal Language Theory and NLP:**

My core research bridges formal language theory and modern natural language processing (NLP). I advance foundational questions in computational theory (e.g., Parikh matrices, language hierarchies, decidability), and transfer these insights to address practical and robust NLP methods. By unifying symbolic, grammar-based approaches with neural and deep learning models, I develop new frameworks for grammar-based data augmentation, few-shot learning, code analysis, and interpretable AI.

- **Unique Contribution:**

Unlike traditional NLP that often relies on black-box neural networks, my work systematically incorporates rigorous mathematical structures—such as Parikh matrix theory and automata—into real-world NLP, code intelligence, and data-centric tasks. This synergy enables provable generalization, robust learning under limited data, and explainable AI.

EDUCATION

Yonsei University

Integrated Ph.D. course in Computer Science

Seoul, KR

2019–Current

- Research focus: computational linguistics, Parikh matrix, formal grammar-based NLP

Yonsei University

Bachelor's degree in Computer Science

Seoul, KR

2015–2019

EXPERIENCE

Theory of Computation Lab, Yonsei University

Research Intern

Seoul, KR

Summer, 2017–Winter, 2018

- Studied automata and formal grammars for language classification tasks
- Explored computational linguistics approaches for simplified NLP tasks

Other Research/Work Experience

2019–Current

- Bridging formal language theory and deep learning for advanced NLP
- Developing grammar-based data augmentation for few-shot NLP tasks
- Investigations on Parikh matrix, M-equivalence, and other language-theoretic properties

PROJECTS

2019–2020: EEG Rule Extraction & Data Similarity Approximation

- Wrote a proposal for the project on EEG stimulation data rule extraction upon entering graduate school in 2019, and led the project.
- Wrote proposals and presentation materials for StarLab funding (2020–2022) on using approximate algorithms for data similarity measurement.

Malware Pattern Extraction & Human-AI Programming Platform Research Project

- Conducted machine learning-based malware pattern extraction and participated in the Human-AI Programming Platform project.
- Researched synergies between NLP and source code, such as code summarization and complexity analysis.

2022–2022: 5th AI Grand Challenge

- Developed a deep learning model for solving mathematical equation problems in a three-stage national-level competition.
- Passed the first stage and wrote the proposal for the competition project.
- Participated in the second-stage competition.

2023–2024: AI Model Learning Trend Analysis for Financial Data

- Collaborated with Bankware Global, leading and planning the research project.
- Analyzed AI models trained on financial data and explored new research directions.

PUBLICATIONS

Under Review

- **MEC³O: Multi-Expert Consensus for Code Time Complexity Prediction**
Joonghyuk Hahn[†], Soohan Lim[†], Yo-Sub Han
Under review
- **A Regex Minimization Benchmark: A PSPACE-Complete Challenge for Language Models**
Hyundong Jin, Joonghyuk Hahn, Yo-Sub Han
Under review
- **Sparsity of M-unambiguous Languages**
Joonghyuk Hahn[†], Yusuke Inoue, Ingyu Baek, Yo-Sub Han
Under review

To Appear

- **Characterizations of M-equivalence and weak M-relation**
Joonghyuk Hahn[†], Hyunjoon Cheon[†], Yo-Sub Han
To appear in International Journal of Foundations of Computer Science (IJFCS) 2025
- **EnCur: Curriculum-Based In-Context Learning with Structural Encoding for Code Time Complexity Prediction**
Joonghyuk Hahn[†], Aditi, Seung-Yeop Baik, Shinwoo Park, Sang-Ki Ko, Yo-Sub Han
To appear in Expert Systems with Applications (ESWA) 2026
- **AmpleHate: Amplifying the Attention for Versatile Implicit Hate Detection**
Yejin Lee, Joonghyuk Hahn, Hyeseon Ahn, Yo-Sub Han
To appear in Proceedings of EMNLP 2025

- **CodeComplex: Dataset for Worst-Case Time Complexity Prediction**
Seung-Yeop Baik, Joonghyuk Hahn, Jungin Kim, Mingi Jeon, Aditi, Yo-Sub Han, Sang-Ki Ko
To appear in Findings of EMNLP 2025

Grammar-Based NLP and AI Applications

- **TCProF: Time-Complexity Prediction SSL Framework**
Joonghyuk Hahn[†], Hyeeseon Ahn, Jungin Kim, Soohan Lim, Yo-Sub Han
In Proceedings of NAACL 2025
- **Advanced Code Time Complexity Prediction Approach Using Contrastive Learning**
Shinwoo Park, Joonghyuk Hahn, Elizabeth Orwig, Sang-Ki Ko, Yo-Sub Han
In Engineering Applications of Artificial Intelligence 2025, Vol. 151
- **SuperST: Superficial Self-Training for Few-Shot Text Classification**
Ju-Hyoung Lee[†], Joonghyuk Hahn[†], Hyeon-Tae Seo, Jiho Park, Yo-Sub Han
In Proceedings of LREC-COLING 2024, 15436–15447
- **GDA: Grammar-based Data Augmentation for Text Classification using Slot Information**
Joonghyuk Hahn[†], Hyunjoon Cheon[†], Elizabeth G. Orwig, Su-Hyeon Kim, Sang-Ki Ko, Yo-Sub Han
In Findings of EMNLP 2023, 7291–7306
- **ATHENA: Mathematical Reasoning with Thought Expansion**
J.B. Kim, Hazel Kim, Joonghyuk Hahn, Yo-Sub Han
In Proceedings of EMNLP 2023, 16315–16327
- **Boosting Code Summarization by Embedding Code Structures**
Jikyeong Son[†], Joonghyuk Hahn[†], Hyeon-Tae Seo, Yo-Sub Han
In Proceedings of COLING 2022, 5966–5977
- **Self-Training using Rules of Grammar for Few-Shot NLU**
Joonghyuk Hahn[†], Hyunjoon Cheon[†], Kyuyeol Han, Cheongjae Lee, Junseok Kim, Yo-Sub Han
In Findings of EMNLP 2021, 4576–4581

Formal Language Theory and Computation

- **Universal Rewriting Rules for the Parikh Matrix Injectivity Problem**
Ingyu Baek[†], Joonghyuk Hahn[†], Yo-Sub Han, Kai Salomaa
In Proceedings of DLT 2024, LNCS 14791, 68–81
- **On the Decidability of Infix Inclusion Problem**
Hyunjoon Cheon[†], Joonghyuk Hahn[†], Yo-Sub Han
In Theory of Computing Systems 2024, Vol. 68, 301–321
- **M-equivalence of Parikh Matrix over a Ternary Alphabet**
Joonghyuk Hahn[†], Hyunjoon Cheon[†], Yo-Sub Han
In Proceedings of CIAA 2023, LNCS 14151, 141–152
- **Most Pseudo-copy Languages Are Not Context-Free**
Hyunjoon Cheon[†], Joonghyuk Hahn[†], Yo-Sub Han, Sang-Ki Ko
In Proceedings of COCOON 2021, 189–200

[†] indicates first author contribution.

TEACHING / MENTORING

Teaching Assistant: Automata and Formal Languages
2019–2022

Yonsei University

- Led discussion sessions and grading assignments in advanced automata theory course.

Mentoring Undergraduate Research Interns
2021–2023

Theory of Computation Lab

- Guided undergraduates on formal grammar concepts and mini-projects in computational linguistics.

PROFESSIONAL ACTIVITIES

Reviewer/PC Member

2022–Current

- Reviewed manuscripts for conferences: e.g. ACL Rolling Review, EMNLP, etc.

SKILLS

- **Programming:** C/C++, Python, Java
- **Tools/Techs:** L^AT_EX, Git, PyTorch, TensorFlow, etc.

REFERENCES

Prof. Yo-Sub Han

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Yonsei University

- Advisor, Department of Computer Science